

Test Plan and Report

CircleHome

Pravin Agrawal-Chung, Cameron Cox, Antonio Wydler, Camille Yabu, Jacqueline

Lopez-Carbajal

May 28, 2026

System Test Scenarios

The following scenarios test user stories. For each scenario, note whether it passed or failed.

Scenario 1: User Registration and Login (Pass / Fail)

User Story: As a new user, I want to create an account so I can use CircleHome.

1. Launch the CircleHome app
2. Select Sign Up; enter email = user@example.com, password = Password123; press Sign Up
3. User should see the home screen or household setup screen
4. Sign out, then select Sign In with the same credentials
5. User should be returned to the home screen successfully

Scenario 2: Create and Join a Household (Pass / Fail)

User Story: As a user, I want to create a household and invite others to join using a code.

1. After login, select Create Household; enter name = Smith House; press Create
2. User should see their household dashboard and a unique invite code
3. On a second device, log in with a different account and select Join Household
4. Enter the invite code from step 2; press Join
5. Second user should appear in the household member list

Scenario 3: Create and Assign a Task (Pass / Fail)

User Story: As a household member, I want to create a task and assign it to a specific member.

1. From the Tasks screen, tap the + button
2. Enter title = Clean kitchen, category = Cleaning, difficulty = 3, due date = 06/01/2027
3. Select a specific household member as the assignee; press Save
4. Task should appear in the task list assigned to the selected member
5. Assigned member should receive a push notification

Scenario 4: Complete and Undo a Task (Pass / Fail)

User Story: As a user, I want to mark a task complete and undo it if needed.

1. From the task list, tap the complete button on an assigned task
2. Task should move to completed; user's leaderboard points should increase
3. Tap the undo button on the completed task
4. Task should return to incomplete status; points should be reversed

Scenario 5: Auto-Assign a Task (Pass / Fail)

User Story: As a user, I want the system to automatically assign a task to the member with the lowest workload.

1. Create a new task and select Auto-Assign instead of choosing a member
2. Press Save
3. Task should be assigned to the member with the lowest current workload score
4. Assigned member should receive a push notification

Scenario 6: Recurring Task Creation (Pass / Fail)

User Story: As a user, I want to create a recurring task that repeats on a schedule.

1. Create a new task; enable the Recurring toggle
2. Set recurrence frequency = Weekly; fill in other fields; press Save
3. Task should appear with a recurring indicator
4. After completion, a new instance should be created for the next recurrence

Scenario 7: Task Swap Request (Pass / Fail)

User Story: As a user, I want to request a task swap with another household member.

1. Select an assigned task and tap Request Swap
2. Select another household member as the swap target; submit the request
3. Target member should receive a swap request notification
4. Target member accepts; task should now be assigned to the target member
5. Repeat but have target member reject; task should remain with original assignee

Scenario 8: Add and View Care Notes (Pass / Fail)

User Story: As a user, I want to log a care note and have all household members see it.

1. Navigate to the Care tab; tap Add Note
2. Enter title = Gave medication, category = Medication, description = 10mg ibuprofen, about = Grandma; press Save
3. Note should appear in the shared care notes list for all household members
4. Attempt to edit the note; the app should not allow editing (notes are immutable)

Scenario 9: Leaderboard and Points (Pass / Fail)

User Story: As a user, I want to see a leaderboard showing members ranked by task completion points.

1. Complete several tasks of varying difficulty levels
2. Navigate to the Leaderboard screen
3. Members should be ranked by total points; higher difficulty should award more points
4. Tap a member's name to view their contribution history

Scenario 10: Profile Setup (Pass / Fail)

User Story: As a user, I want to set my display name and profile photo.

1. Navigate to Profile settings
2. Enter display name = Alice; upload a profile photo; set workload level; press Save
3. Profile photo and name should update across the app
4. Existing task assignments should reflect the updated name

Unit Tests

All automated tests are located at:

frontend/test/

app_state_test.dart — AppState.initialize

- Clears data when initialized with null user — Pass / Fail
- Loads user data when initialized with a valid user — Pass / Fail
- Loads household data when user belongs to a household — Pass / Fail
- Leaves householdData null when user has no householdId — Pass / Fail
- uid and email getters return correct values — Pass / Fail

app_state_test.dart — AppState.joinHouseholdByCode

- Returns false and sets error for invalid code — Pass / Fail
- Returns true and updates state for valid code — Pass / Fail
- Returns false when uid is null — Pass / Fail

app_state_test.dart — AppState.taskStream

- Returns null when no household is set — Pass / Fail
- Returns a stream when household is loaded — Pass / Fail

auth_service_test.dart — AuthService

- currentUser is null when not signed in — Pass / Fail
- currentUser is set after sign in — Pass / Fail
- signUp creates a new user — Pass / Fail
- signOut clears the current user — Pass / Fail
- authStateChanges emits null when signed out — Pass / Fail
- authStateChanges emits user after sign in — Pass / Fail

firestore_service_test.dart — addTask

- Creates a task document with correct fields — **Pass** / Fail
- Stores correct difficulty value — **Pass** / Fail
- Stores recurring task with frequency — **Pass** / Fail

firestore_service_test.dart — completeTask

- Marks task as completed — **Pass** / Fail
- Creates an activity entry — **Pass** / Fail

firestore_service_test.dart — uncompleteTask

- Sets completed back to false — **Pass** / Fail

firestore_service_test.dart — deleteTask

- Removes the task document — **Pass** / Fail
- Does not affect other tasks — **Pass** / Fail

firestore_service_test.dart — addCareNote

- Creates a care note with correct fields — **Pass** / Fail

firestore_service_test.dart — deleteCareNote

- Removes the care note document — **Pass** / Fail

firestore_service_test.dart — joinHouseholdByCode

- Returns false for an invalid code — **Pass** / Fail
- Returns true and sets role to member for a valid code — **Pass** / Fail
- Adds householdId to user householdIds list — **Pass** / Fail

firestore_service_test.dart — getCareNotes

- Returns only notes for the given household — **Pass** / Fail
- Returns empty when household has no notes — **Pass** / Fail

firestore_service_test.dart — getTasks

- Returns only incomplete tasks for the household — **Pass** / Fail

firestore_service_test.dart — autoAssignTask

- Returns null when household has no members — **Pass** / Fail
- Assigns to the member with the lower task load — **Pass** / Fail
- Assigns to single available member — **Pass** / Fail

- Skips high-workload members for hard tasks — **Pass** / Fail
- Falls back to all members if everyone is high-workload on hard task — **Pass** / Fail
- Weights overdue tasks more heavily — **Pass** / Fail

task_utils_test.dart — difficultyLabel

- Returns Very Easy for 1 — **Pass** / Fail
- Returns Easy for 2 — **Pass** / Fail
- Returns Moderate for 3 — **Pass** / Fail
- Returns Hard for 4 — **Pass** / Fail
- Returns Very Hard for 5 — **Pass** / Fail
- Returns empty string for out-of-range value — **Pass** / Fail

task_utils_test.dart — difficultyColor

- Returns green for difficulty 1 — **Pass** / Fail
- Returns orange for difficulty 3 — **Pass** / Fail
- Returns red for difficulty 5 — **Pass** / Fail
- Clamps to valid range for 0 — **Pass** / Fail
- Clamps to valid range for 6 — **Pass** / Fail

task_utils_test.dart — isTaskOverdue

- Returns true for a date in the past — **Pass** / Fail
- Returns false for a date in the future — **Pass** / Fail
- Uses provided now parameter — **Pass** / Fail
- Not overdue when due is after now — **Pass** / Fail

time_utils_test.dart — formatRelativeTime

- Shows seconds for times under a minute ago — **Pass** / Fail
- Shows minutes for times under an hour ago — **Pass** / Fail
- Shows hours for times under a day ago — **Pass** / Fail
- Shows days for times under a week ago — **Pass** / Fail
- Shows weeks for times over a week ago — **Pass** / Fail
- Rounds down for partial weeks — **Pass** / Fail
- Shows 59s ago for 59 seconds — **Pass** / Fail
- Shows 1m ago at exactly 60 seconds — **Pass** / Fail

widget_test.dart — Widget Tests

- App launches via MyApp() — **Pass** / Fail
- Scaffold renders body content — **Pass** / Fail
- NavigationBar shows correct labels (Home, Tasks, Care) — **Pass** / Fail
- CircularProgressIndicator visible during loading — **Pass** / Fail
- Empty state shows No care notes yet and Add First Note button — **Pass** / Fail

